MIL-S-20560
29 December 1951

MILITARY SPECIFICATION

SHELL, STEEL FOR SMOOTH-BORE MORTARS, WELDED TYPE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-37-2
10 December 1943

This specification consists of this cover sheet, Specification 50-37-2, dated 10 December 1943, and Amendment 3, dated 5 April 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 50-37-2
Amendment 3
5 April 1949
Superseding
Amendment 2

SHELL, STEEL FOR SMOOTH-BORE MORTARS, WELDED TYPE

The following changes shall be made in Specification No. 50-37-2, dated 10 December 1943:

Page 1, paragraph C-5, delete the paragraph and substitute the following:
"C-5. The interior cavity of each shell shall be cleaned by the shot or snad blast method, or by any other method approved by the Chief of Ordnance."

Page 1, paragraph C-6, delete the paragraph and substitute the following:
"C-6. Cleaning Prior To Painting.--The shell surfaces to be painted shall be cleaned and prepared in accordance with Grade I, Specification JAN-C-490. These cleaned surfaces shall not be recontaminated prior to painting. After painting, these surfaces shall meet the various performance requirements of Specification JAN-C-490."

Page 2, paragraphs D-5, D-5a and D-5b, delete these paragraphs entirely.

Page 4, paragraph F-7. Delete the paragraph and substitute the following:
"F-7. Repairs and Retests.
F-7a. Repairs by Welding. A shell showing leaks or welding defects (such as lack of fusion, etc.) may be repaired by arc welding (shielded arc). No single leak exceeding 1/-inch in length shall be repaired and not over 2 inches of defective welding may be repaired in any one shell. The flaw or defect shall be completely removed before welding and a weld of 100% penetration shall be required. However, if the flaw of defect is less than 1/3 the thickness of the shell, a weld of 100% penetration will not be necessary. The repaired shell shall then be tested at 4,000 lbs. per square inch min. hydraulic pressure. Any shell failing in this retest shall be finally rejected.
F-7b. A shell with scars or pits within the welded areas, but not in the bourrelet, which were caused by lack of fusion need not be repaired by welding if the depth of such imperfection is not greater than 1/3 the thickness of the shell wall and their size does not extend beyond the limits of 1/2 inch in lenght of 1/8 inch in width, and that the number of such imperfections does not exceed three with a minimum spacing of 1/4 inch. Shell in which the depth of the imperfections cannot easily be ascertained must be repaired and tested in accordance with paragraph F-7a in order to be considered for acceptance. All shell with imperfections which fall within the stated limits shall be tested at 4,000 lbs. per square inch minimum hydraulic pressure. Any shell failing either through leakage or distortion shall be finally rejected. All imperfections in shell that are acceptable under these provisions be filled with an approved paint filler prior to painting of the shell.
F-7c. All shell with imperfections within the limits permitted by paragraph F-7b which also meet the hydraulic pressure test requirements shall be marked with a suitable stamp on the ogive of the shell. (See Fig. 1)."
Add figure 1.

The purpose of this amendment is to add Figure 1 inadvertently omitted from Amendment -2.

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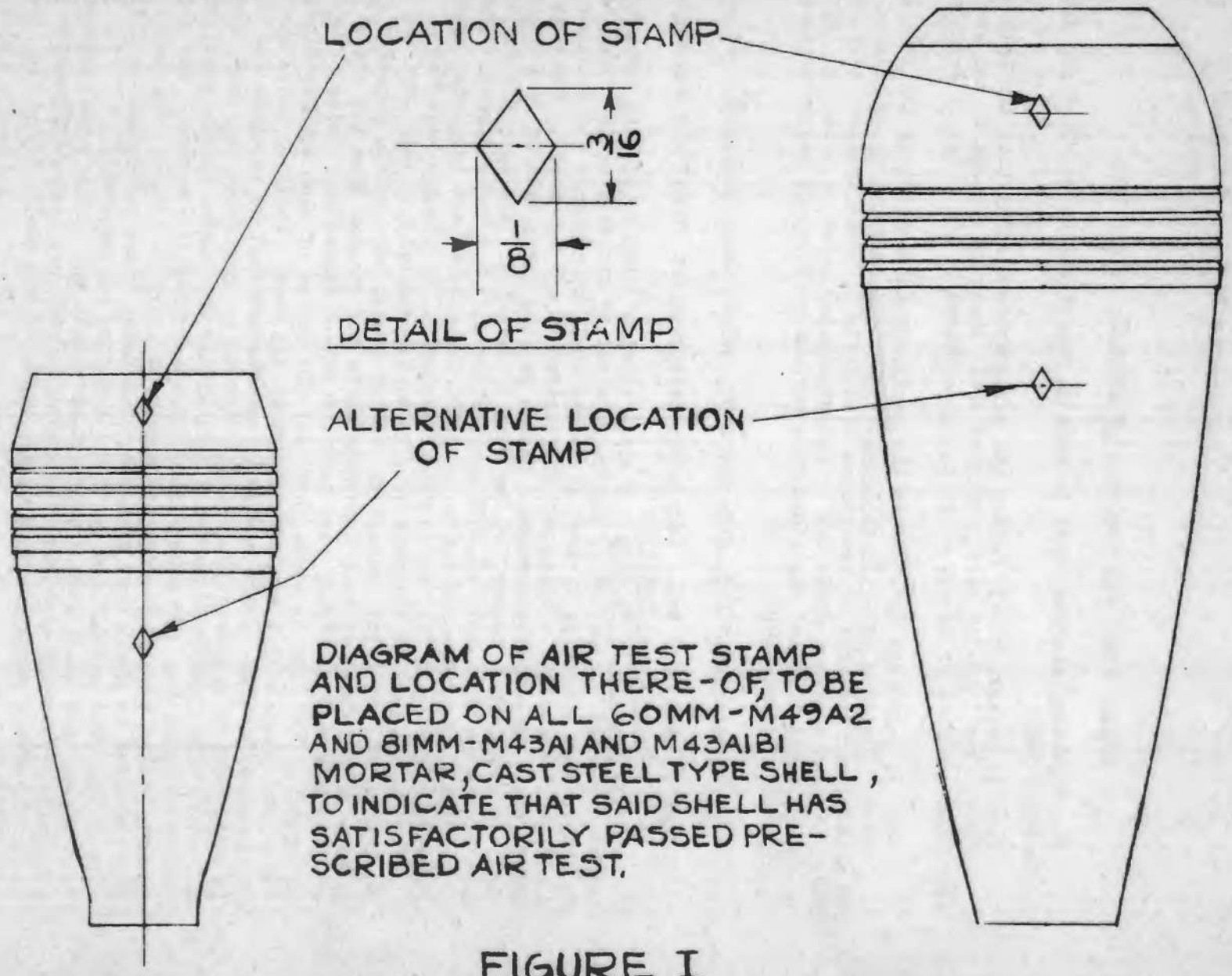


DIAGRAM OF AIR TEST STAMP AND LOCATION THERE-OF, TO BE PLACED ON ALL 60MM-M49A2 AND 81MM-M43A1 AND M43A1B1 MORTAR, CAST STEEL TYPE SHELL, TO INDICATE THAT SAID SHELL HAS SATISFACTORILY PASSED PRESCRIBED AIR TEST.

FIGURE I

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U. S. ARMY
SPECIFICATION

No. 50-37-2
10 December 1943

SHELL, STEEL FOR SMOOTH-BORE MORTARS, WELDED TYPE

A. APPLICABLE SPECIFICATIONS

A-1. The Ordnance Department drawings pertaining to the shell under contract, and all drawings and specifications listed or referred to thereon, form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPES

B-1. This specification covers high explosive, chemical, and practice shell for smooth-bore mortars, manufactured by the butt resistance welding process..

C. MATERIAL AND WORKMANSHIP

C-1. Material and parts shall be in accordance with applicable specifications and drawings.

C-2. The steel used shall be manufactured by the open-hearth or electric furnace process.

C-3. The chemical composition of the steel shall conform to the requirements of W.D. 1020.

C-4. The steel shall be of uniform quality and condition and free from seams, tears, laminations, slivers or other injurious defects.

C-5. Upon completion of the machining operations, the shell shall be cleaned to free them of all grease, dirt, chips, and other foreign material, preparatory to painting.

C-6. The interior and exterior coatings specified on the drawings, for the shell under contract shall be applied in such manner as to insure an even distribution and a thorough coating. The shell shall be stencilled or marked as prescribed on the drawing.

C-7. Process qualification. Before quantity production on an order is started, the contractor shall submit 10 shells made in accordance with the process methods he proposes to use. These shell will be tested as specified in paragraph F-3.

D. GENERAL REQUIREMENTS

D-1. Shell may be made by welding together parts formed either by forging or by stamping, or by both forging and stamping.

D-2. The grain of the metal in the body of the shell shall run in a substantially circumferential direction.

D-3. Welding.

D-3a. Exclusive of the welding of one interior diaphragm, not more than two longitudinal welds or more than one transverse weld shall be used.

D-3b. Either upset or flash butt resistance welding may be used. The diaphragm in cartridge container end of shell may be welded to the body or preferably formed integral with it.

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D-3c. The weld shall be completely fused. The fusion zone shall be free from slag, scale, porosity, gas pockets and other defects which would appreciably affect strength of the welded joint.

D-3d. All metal burrs shall be removed except that in the interior of the shell a portion of the longitudinal bead of regular contour and not exceeding .01 inch in height will be permitted.

D-3e. The resistance welding equipment employed shall be of ample capacity to secure optimum properties of the welds. The equipment shall possess such sensitive control of current, pressure, and time elements as will permit of uniform welding of highest quality.

D-4. Soundness.

D-4a. Finished shell shall withstand an internal hydrostatic pressure of 4,000 pounds per square inch without deformation or evidence of cracks and leakage.

D-4b. Shells that fail or show leaks in hydraulic tests shall be tentatively rejected. (See paragraph F-7).

D-5. Cleaning.

D-5a. The interior cavity of each shell shall be cleaned by the shot or sand-blast method, or by a method approved by the Chief of Ordnance.

D-5b. The exterior of the shell may be cleaned by any method satisfactory to the inspector.

E. DETAIL REQUIREMENTS

E-1. The shell shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Physical properties. The steel used in the manufacture of the shell body shall have the following physical properties: (See paragraph F-6).

Yield point	- 35,000 lbs. per square inch, minimum.
Elongation in 2 inches	- Not less than 15 percent.
Reduction in area	- Not less than 30 percent.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection. Inspection shall be as prescribed in U. S. Army Specification No. 50-0-1, "General Specifications for Ammunition Except Small Arms Ammunition," and Federal Specification No. QQ-M-151, "Metals, General Specification for Inspection of."

F-2. Lots.

F-2a. The maximum size of the lots shall be 20,000. A smaller quantity may constitute a lot at the discretion of the inspector.

F-2b. The first lot manufactured on any contract shall not exceed 5,000 shell, except when the new contract immediately follows the manufacture of satisfactory material of the same character.

F-3. Process qualification tests. The ten sample shell submitted for process qualification tests shall be tested as follows:

F-3a. Six shell shall be given a hydraulic test at an internal pressure at which the metal of the body is stressed up to the prescribed yield point of the material which shall be a minimum of 35,000 pounds per square inch. If failure of the metal occurs in the weld zone, the processing methods shall be rejected. If failure occurs due to a defect outside the weld zone, a retest shall be allowed.

F-3b. After the hydraulic pressure test, all shell shall be given an air

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pressure test at 300 pounds per square inch. All shell tested shall be free from air leakage.

F-3c. Three shell shall be sectioned across each of the welds. Samples containing each weld shall be given a macro examination. Under a magnification of 10 the welds shall be completely fused and free of discontinuities and harmful defects. The metal shall be free of hard spots and other non-homogeneities which might decrease machinability.

F-3d. Flattening test.- A section not less than 2 inches long taken from middle section of shell shall stand flattening between parallel plates, until the distance between the plates is seven times the wall thickness, without showing welding flaws, such as lack of fusion, etc. The weld shall be placed in the line of maximum bend. Three shell shall be tested in this manner.

F-3e. In case it is questionable whether any shell or sample thereof meets the requirements of these specimens, a micro examination of the shell or speciment shall be made to determine the nature and extent of defects.

F-3f. In case of failure to meet all these requirements, a complete retest shall be made.

F-3g. In case of change in important features of processing methods or of materials, a requalification of process shall be required.

F-4. Hydraulic pressure test. Each welded shell shall be tested at 4,000 pounds per square inch hydraulic pressure after machining is completed.

F-5. Chemical analysis. At least two samples for check chemical analysis shall be selected by the inspector from the material from each heat used to form the shell and at least one sample shall be selected from each lot of 25 tons or fraction thereof from each heat of metal used.

F-6. Test for Physical properties.

F-6a. At the option of the contractor, the determination of physical properties of the steel by means of tension test specimens as required by paragraph E-2, may be eliminated and a process control test substituted, subject to the provisions governing failures in the process control test.

F-6b. For the process control test, ten (10) shell shall be taken from each heat of steel, suitably marked for later identification by the inspector, and processed through the complete sequence of the blanking, forming, welding and machining operations (with allowable exception of final grinding of the bourrelet, and tapping of nose and base threads). The shell shall be subjected to a hydraulic test at an internal pressure of 6,500 pounds per square inch in the presence of the inspector followed by a careful inspection by the inspector to ascertain if any distortion has taken place in the shell body.

F-6c. Distortion shall be measured at a distance from the base of the shell equivalent to 40 percent of the distance between the base of the shell and the rear edge of the bourrelet, and shall be measured diametrically at a position approximately 90° removed from the weld zone. Distortion shall be defined as an increase in diameter produced by the hydraulic test, which exceeds .005 inch at this position after the load has been removed. The contractor shall machine suitable flat areas on the shell body at the positions where the measurements are to be taken. Approval of these shell by the inspector shall be considered as approval of the represented heat of steel for processing.

F-6d. Failure of any one of the test shell in the hydraulic pressure test, as indicated by distortion beyond the limit specified in paragraph F-6c, shall require the determination of the physical properties of the rolled plate by means of tension tests, subject to the provisions of paragraph F-6e. Acceptance

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of the represented heat of rolled plate upon failure in the hydraulic pressure test shall be governed by the material meeting the physical properties required by paragraph E-2.

F-6e. Failure of the metal in the weld zone of any test shell shall require that double the number of shell which failed be selected from the heat and processed as required by paragraph F-6b. No tension tests will be required if failure occurs in the weld. However failure of the metal outside the weld zone, or together with failure in the weld zone, shall require that twenty (20) additional shell be processed from the represented heat and subjected to the hydraulic pressure test, as well as require tension tests obtained from the rolled plate. Failure of the metal in any one of the twenty shell outside the weld zone or distortion beyond the .005 inch limit, or failure of any tension test to meet the required physical properties, shall be cause for rejection of the heat.

F-7. Repairs and retests. A shell showing leaks or welding defects (such as lack of fusion, etc.) may be repaired by arc welding (shielded arc). No single leak exceeding 1/2-inch in length shall be repaired and not over 1 inch of defective welding may be repaired in any one shell. The flaw or defect shall be completely removed before welding and a weld of 100 percent penetration shall be required. However, if the flaw or defect is less than 1/3 the thickness of the shell, a weld of 100 percent penetration will not be necessary. The repaired shell shall then be tested at 4,000 pounds per square inch hydraulic pressure. Any shell failing on this retest shall be finally rejected.

F-8. There shall be installed in each plant engaged in machining shell, one set of hydraulic equipment capable of testing shell by an internal hydrostatic pressure to destruction. The equipment shall be used by the inspector for testing doubtful shell, such as those which show evidence of cracks or porosity.

F-9. In determining the hydraulic pressure to be used for stressing the metal of the body up to the prescribed yield point of the material, the following formula shall be used:

$$P = \frac{2\,St}{D}$$

Where P = Pressure in pounds per square inch.
S = Fiber stress.
t = Thickness of wall.
D = Outside diameter in inches.

G. PACKAGING, PACKING, AND MARKING

G-1. Packing.

G-1a. Shells shall be packed in commercial type containers so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

G-1b. Data card. A data card, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted shell in accordance with the form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket and six cards completely filled out shall be furnished the inspector. Lot numbers of contents shall be marked plainly on each container.

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AMMUNITION DATA CARD

Kind_____ Ammunition Lot No._____
Quantity in lot_____
Drawing No._____ Rev._____ Spec. No._____ Rev._____
Manufactured by_____ Date_____
Contract No._____ Order No._____
Packed_____ in a box.

Remarks_____

Certified to by_____ Inspector
(Print here name of inspector, district, arsenal, or loading plant, and sub-district where applicable)

G-2. Marking. The marking for shipment shall be in accordance with U.S. Army Standard Specifications for Marking Shipment, 100-2.

H. NOTES

H-1. The shell covered by this specification are for use in Smooth-Bore Mortars.

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the U. S. Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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